

Module – 1

Android

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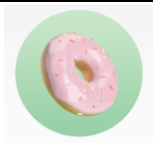
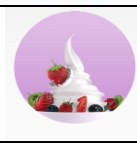
Android

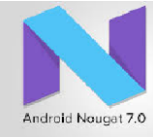
- Android is an open source and Linux-based operating system for mobile devices such as smartphones and tablet computers.
- Designed primarily for touchscreen mobile devices such as smartphones and tablets.
- Android was developed by the Open Handset Alliance, led by Google
- Android programming is based on Java programming language
- Android offers a unified approach to application development for mobile devices which means developers need only develop for Android, and their applications should be able to run on different devices powered by Android.
- Android's user interface is mainly based on direct manipulation, using touch gestures that loosely correspond to real-world actions, such as swiping, tapping and pinching, to manipulate on-screen objects, along with a virtual keyboard for text input.
- In addition to touchscreen devices, Google has further developed Android TV for televisions, Android Auto for cars, and Android Wear for wrist watches, each with a specialized user interface.
- Variants of Android are also used on game consoles, digital cameras, PCs and other electronics.
- Android Package Kit (APK) is the package file format used by the Android operating system for installation of mobile app
- Android applications ("apps") can be downloaded from the Google Play store
- Android has been the best-selling OS on tablets since 2013, and runs on the vast majority of smartphones. As of May 2017, Android has two billion monthly active users, and it has the largest installed base of any operating system.
- Android's source code is released by Google under an open source license
- Android is popular with technology companies that require a ready-made, low-cost and customizable operating system for high-tech devices.
- Android Studio and Eclipse is the official integrated development environment (IDE) for Google's Android operating system for Android development.
- Android Studio and Eclipse are available for download on Windows, macOS and Linux based operating system.
- The first commercially available smartphone running Android was the HTC Dream, also known as T-Mobile G1, announced on September 23, 2008.

1. History

2003	Android Inc. was founded
2004	Company were to develop an advanced operating system for digital cameras
2005	Google acquired Android Inc.
2008	first commercially available smartphone running Android was the HTC Dream, also known as T-Mobile G1
2010	Google launched its Nexus series of devices
2014	Google announced Android One
2016	Google introduced the Pixel and Pixel XL smartphones

2. Version

Year	Version Number	Code Name	API	Logo
2008	1.0	No Code Name	1	-----
2009	1.1	No Code Name	2	-----
2009	1.5	Cupcake	3	
2009	1.6	Donut	4	
2009	2.0 – 2.1	Eclair	5-7	
2010	2.2 – 2.2.3	Froyo	8	
2010	2.3 – 2.3.7	Gingerbread	9-10	

2011	3.0 – 3.2.6	Honeycomb	11-13	
2011	4.0 – 4.0.4	Ice Cream Sandwich	14-15	
2012	4.1 – 4.3.1	Jelly Bean	16-18	
2013	4.4 – 4.4.4	KitKat	19-20	
2014	5.0 – 5.1.1	Lollipop	21-22	
2015	6.0 – 6.0.1	Marshmallow	23	
2016	7.0 - 7.1.2	Nougat	24-25	
2017	8.0 – 8.1.0	Oreo	26-27	

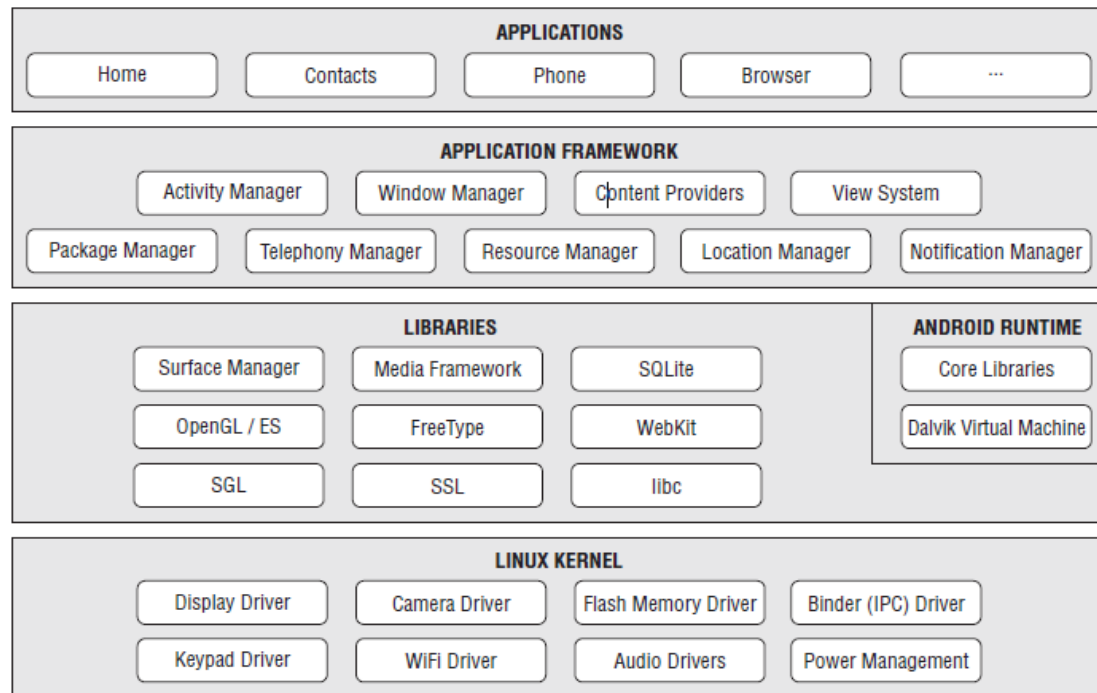
3. Features of Android

- **Storage** — Uses SQLite, a lightweight relational database, for data storage.
- **Connectivity** — Supports GSM/EDGE, IDEN, CDMA, EV-DO, UMTS, Bluetooth, WiFi, LTE, and WiMAX.
- **Messaging** — Supports both SMS and MMS.
- **Web browser** — Based on the open-source WebKit, together with Chrome's V8 JavaScript engine
- **Media support** — Includes support for the following media: H.263, H.264 (in 3GP or MP4 container), MPEG-4 SP, MP3, WAV, JPEG, PNG, GIF, and BMP
- **Hardware support** — Accelerometer Sensor, Camera, Digital Compass, Proximity Sensor, and GPS
- **Multi-touch** — Supports multi-touch screens
- **Multi-tasking** — Supports multi-tasking applications

- **Flash support** — Android 2.3 supports Flash 10.1.
- **Tethering** — Supports sharing of Internet connections as a wired/wireless hotspot

4. Architecture of Android

In order to understand how Android works, take a look at Figure, which shows the various layers that make up the Android operating system (OS).



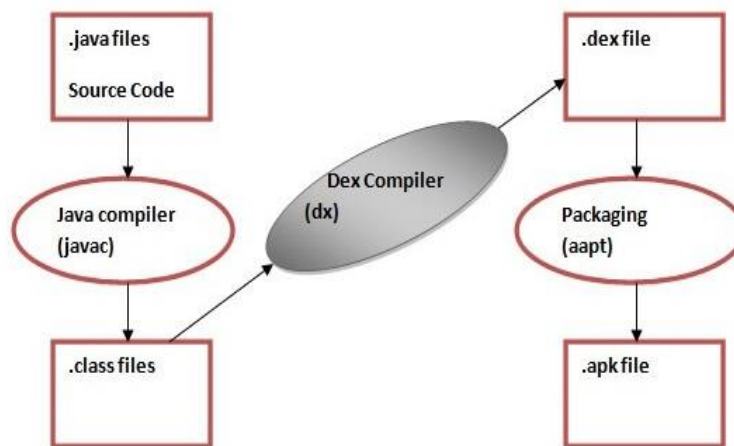
The Android OS is roughly divided into five sections in four main layers:

- **Linux kernel** — This is the kernel on which Android is based. This layer contains all the low-level device drivers for the various hardware components of an Android device.
- **Libraries** — These contain all the code that provides the main features of an Android OS. For example, the SQLite library provides database support so that an application can use it for data storage. The WebKit library provides functionalities for web browsing.
- **Android runtime** — At the same layer as the libraries, the Android runtime provides a set of core libraries that enable developers to write Android apps using the Java programming language. The Android runtime also includes the Dalvik virtual machine, which enables every Android application to run in its own process, with its own instance of the Dalvik virtual machine. Dalvik is a specialized virtual machine designed specifically for Android and optimized for battery-powered mobile devices with limited memory and CPU.
- **Application framework** — Exposes the various capabilities of the Android OS to application developers so that they can make use of them in their applications.

- **Applications** — At this top layer, you will find applications that ship with the Android device (such as Phone, Contacts, Browser, etc.), as well as applications that you download and install from the Android Market. Any applications that you write are located at this layer.

Dalvik Virtual Machine | DVM

- The **Dalvik Virtual Machine (DVM)** is an android virtual machine optimized for mobile devices.
- It optimizes the virtual machine for *memory*, *battery life* and *performance*.
- Dalvik is a name of a town in Iceland. The Dalvik VM was written by Dan Bornstein.
- The Dex compiler converts the class files into the .dex file that run on the Dalvik VM. Multiple class files are converted into one dex file.
- Let's see the compiling and packaging process from the source file:



- The javac tool compiles the java source file into the class file.
- The dx tool takes all the class files of your application and generates a single .dex file. It is a platform-specific tool.
- The Android Assets Packaging Tool (aapt) handles the packaging process.

Advantages

- DVM supports the Android operating system only.
- In DVM executable is APK.
- Execution is faster.
- From Android 2.2 SDK Dalvik has it's own JIT (Just In Time) compiler.
- DVM has been designed so that a device can run multiple instances of the Virtual Machine effectively.
- Applications are given their own instances.

5. **Android Devices in the Market**

Android devices come in all shapes and sizes.

- Smartphones
- Tablets
- Android Auto
- Android Wear
- E- Netbooks
- MP4 players
- Internet TVs
- reader devices
- Digital Camera

6. **Android Software Development Kit (SDK)**

- The Android SDK provides you the API libraries and developer tools necessary to build, test, and debug apps for Android.
- The Android Software Development Kit (Android SDK) provides all necessary tools to develop Android applications.
- Android SDK include a debugger, libraries, a handset emulator, documentation, sample code, and tutorials.
- The Android SDK contains tools to create, compile and package Android application.
- The Android SDK also provides an Android device emulator, so that Android applications can be tested without a real Android phone.
- We can create Android virtual devices (AVD) via the Android SDK, which run in this emulator.
- Currently supported development platforms include computers running Linux, Mac OS and Windows.

7. **Android Development Tools (ADT)**

- Android SDK Tools is a component for the Android SDK. It includes the complete set of development and debugging tools for Android. It is included with Android Studio.
- ADT extends the capabilities quickly set up new Android projects, create an application UI, add packages based on the Android Framework API, debug your applications using the Android SDK tools.

8. **List Various Mobile Technologies**

- Android

- Apple IOS
- Windows
- Symbian
- BlackBerry OS
- webOS or Palm OS
- BADA;
- Maemo
- MeeGo
- Ubuntu Touch

9. Compare IOS and Android

	Android	IOS
Developer	Various, mostly Google and Open Handset Alliance	Apple Inc.
Initial release	September 23, 2008	July 29, 2007
Latest stable release and Updates	Android 12	iOS 15 and iPadOS 15
Customizability	A lot. Can change almost anything.	Limited unless jailbroken
Source model	Open source	Closed, with open source components.
File transfer	Easier than iOS. Using USB port and Android File Transfer desktop app. Photos can be transferred via USB without apps.	More difficult. Media files can be transferred using iTunes desktop app. Photos can be transferred out via USB without apps.
Internet browsing	Google Chrome (other browsers are available). Any browser app can be set as default. Ad blocking is supported with Firefox.	Safari. Any browser app can be set as default but they all use the same rendering engine (Safari/Webkit) behind the scenes. Ad blocking is supported via content blockers like Firefox Focus.
Web mapping service	Google Maps	Apple Maps (default). Google Maps also available via a separate app download, but not as default.
Available language(s)	100+ languages	40 languages

Video chat	Google Meet and other 3rd party apps	FaceTime (Apple devices only) and other 3rd party apps
Virtual assistant	Google Assistant	Siri
Available on	Many phones and tablets . Major manufacturers such as Samsung, Oppo, OnePlus, Vivo, Honor and Xiaomi. Android One devices are pure Android. Pixel line of devices is made by Google, using a almost pure version of Android	iPod Touch, iPhone, iPad, Apple TV (2nd and 3rd generat

10. Eclipse IDE

- One of the integrated development environment (IDE) for developing your Android applications IDE is Eclipse.
- Eclipse IDE is a multi-language software development environment.
- It can be used to develop several types of applications, using languages such as Java, Ada, C, C++, COBOL, Python, and others.
- Eclipse is a platform independent, means available in Windows, Mac OS X, and Linux.
- Eclipse software development kit (SDK) is free and open-source software
- Eclipse supports development for Tomcat, GlassFish and many other servers

Install eclipse in Windows

1. Download the Eclipse Installer. Download Eclipse Installer from <http://www.eclipse.org/downloads>.
2. Start the Eclipse Installer executable.
3. Select the package to install.
4. Select your installation folder.
5. Launch Eclipse.

Features

- a) Automated Error Reporting
- b) Platform Improvements
- c) Free and Open Source
- d) Most used Java IDE
- e) Syntax Checking
- f) Refactoring
- g) Code Completion

h) Extensions and Plugins

11. Android studio IDE

- Android Studio is the official integrated development environment (IDE) for Google's Android operating system
- designed specifically for Android development
- It is available for download on Windows, macOS and Linux based operating systems
- License is open source

Installation of Android Studio on Windows

To download Android Studio for your computer, you need to visit official site of android. There you will see android studio download button.

After downloading android studio, follow below steps to install android studio on your computer.

Step1: Open you downloaded android studio file and wait for few seconds, it will take some time to load on your screen.

Step2: After that the window will appear on your screen, where it will show you welcome to android studio. Go for "**Next**"

Step3: Now choose what you want to install, all are most important component in android programming. Go for "Next"

Step4: Go for "I Agree", It's for integrating SDK, it will confirm SDK installation.

Step5: Go for "I Agree", It's for Haxm installation.

Step6: Now choose the location for Android Studio and SDK.
Go for "Next"

Step7: Here you need to define the size of android emulator processor. You can define up to 2GB and more than that, depends on your RAM capacity. The size will be determined through Haxm. This will help your android studio to run faster.

Step8: Now choose setting for shortcut of android studio on your computer.

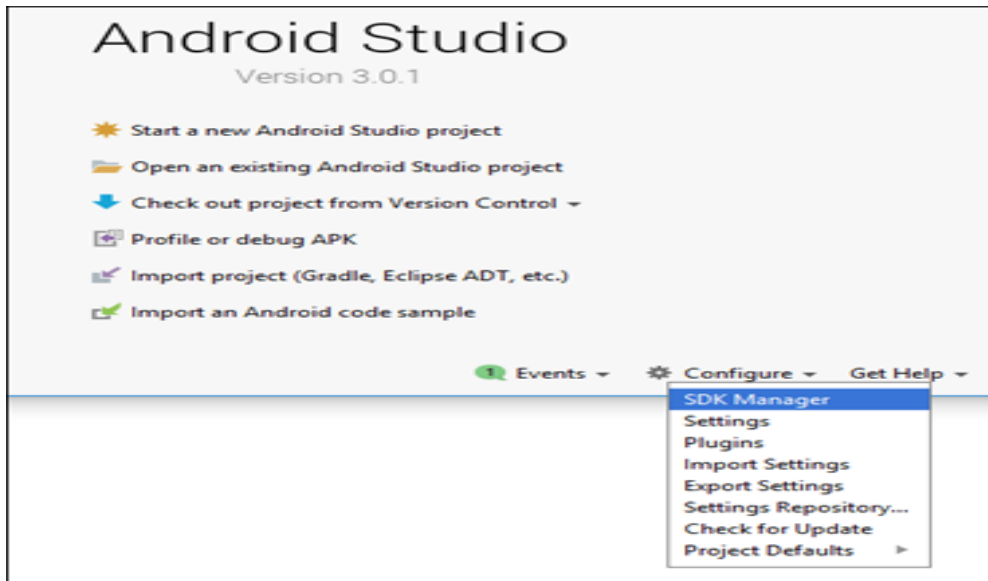
Step9: After that it will take lots of time to copy all the files to the storage location you have selected before.

Step10: click next, when finish appears click on Finish.

Installation of SDK in android studio

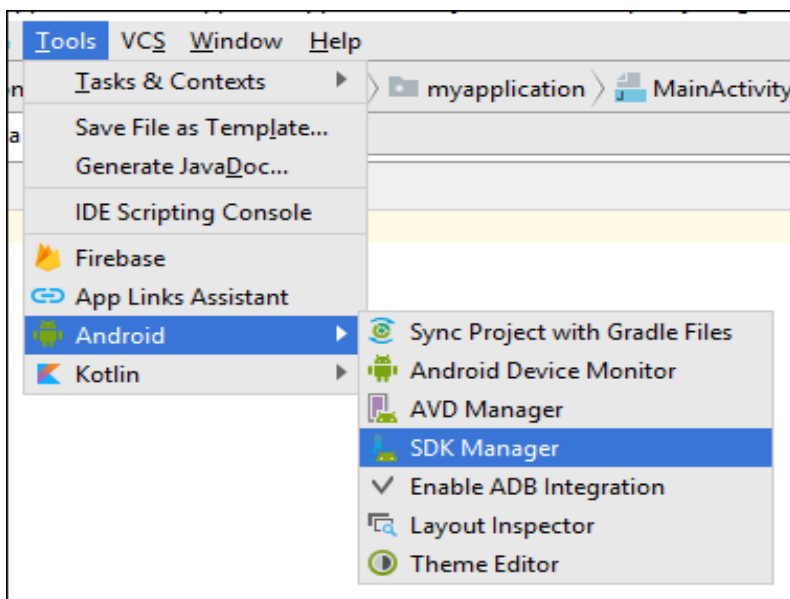
Step 1: Start Android Studio.

Step 2: To open SDK Manager, do any of the these: On Android Studio landing page, select Configure > SDK Manager. ...



(OR)

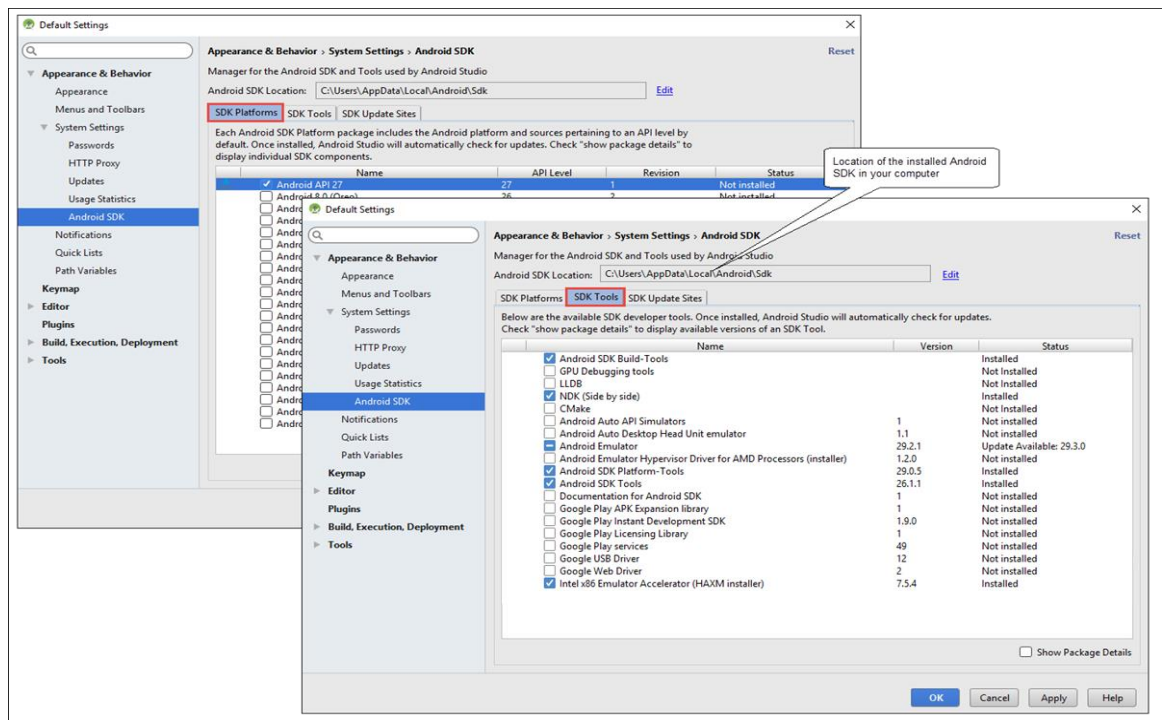
- From your Android Studio application toolbar, select **Tools** > **Android** > **SDK Manager**.



Step 3: In the Default Settings dialog box, click these tabs to install Android SDK platform packages and developer tools. ..

Step 4: In the SDK Tools tab, select Android SDK Build-Tools 30(or above).

Step 5: Click OK to install the SDK.



Features of Android studio

1.Optimized for all Android devices

- Android Studio provides a unified environment where you can build apps for Android phones, tablets, Android Wear, Android TV, and Android Auto.
- Structured code modules allow you to divide your project into units of functionality that you can independently build, test, and debug.

2.Instant App Run

- It is an advanced technology in which it cleverly understands the transmutations done in the applications and deliver it instantly without taking time to rebuild the apk and installations make.

3. Visual Layout Editor

- Layout editor helps to build the layout quickly by adding different attributes either by hard-code or drag and drop.
- The preview of the codes can be seen easily on the visual editor screen and changes can be made accordingly by resizing it dynamically.
- This will make testing the application process more facile and more exhaustive.

4. Fast Emulator

- Android has a great feature of Emulator which is exactly like the android phones to test how the application looks like in physical devices.
- It gives real-time experience to the Android applications.
- It allows you to test your applications faster and on different-different configuration devices like tablet, android phone etc.

5. Intelligence Code Editor

- Android Studio provides you with the intelligent and quick code editor.
- This will help you and guide you with the accurate code.
- Android studio has the special feature of the development of code by the drop-down list with suggesting the code you can integrate.

6 . Addition of New Activity as a Code Template

- Yes, Android also has the feature of templates built-in. If you know to build that accordingly that makes your task easier.

8. Help to Connect with Firebase

- Android Studio helps to give real-time experience with IOT based project development with dynamic upgrades in the application.
- Firebase connectivity help to create direct updates and provide databases connectivity.

9. Support KOTLIN

- Kotlin the official language can be considered for Android.
- It is a language without having any new restrictions and has various advantages in it.
- The great feature of Kotlin is it run without any disturbances in older android versions as well that means no issues of specific android versions.
- It runs fast and equivalent to Java.
- Java developers can easily hands-on Kotlin with no-issues because it based on automated Java only.

10.Colour Previews

- Android studio helps to see the code XML part in a preview to know that how perfectly we are designing the application according to the need before launching the application.
- It provides powerful functionality and enhanced features of drag and drops or resizes the application.

12. Android Virtual Device (AVD)

- AVD stands for *Android Virtual Devices*.
- An AVD is an emulator instance that enables you to model an actual device.
- Each AVD consists of a hardware profile, a mapping to a system image, as well as emulated storage, such as a secure digital (SD) card.
- You can create as many AVDs as you want in order to test your applications with several different configurations.
- This testing is important to confirm the behavior of your application when it is run on different devices with varying capabilities.
- An Android Virtual Device (AVD) is a configuration that defines the characteristics of an Android phone, tablet, Android Wear, or Android TV device that you want to simulate in the Android Emulator.
- The AVD Manager is an interface you can launch from Android Studio that helps you create and manage AVDs.

13. Create Android Virtual Device (AVD)**Step 1:**

- To open the AVD Manager, do one of the following:

- Select Tools -> Android -> AVD Manager. (OR)

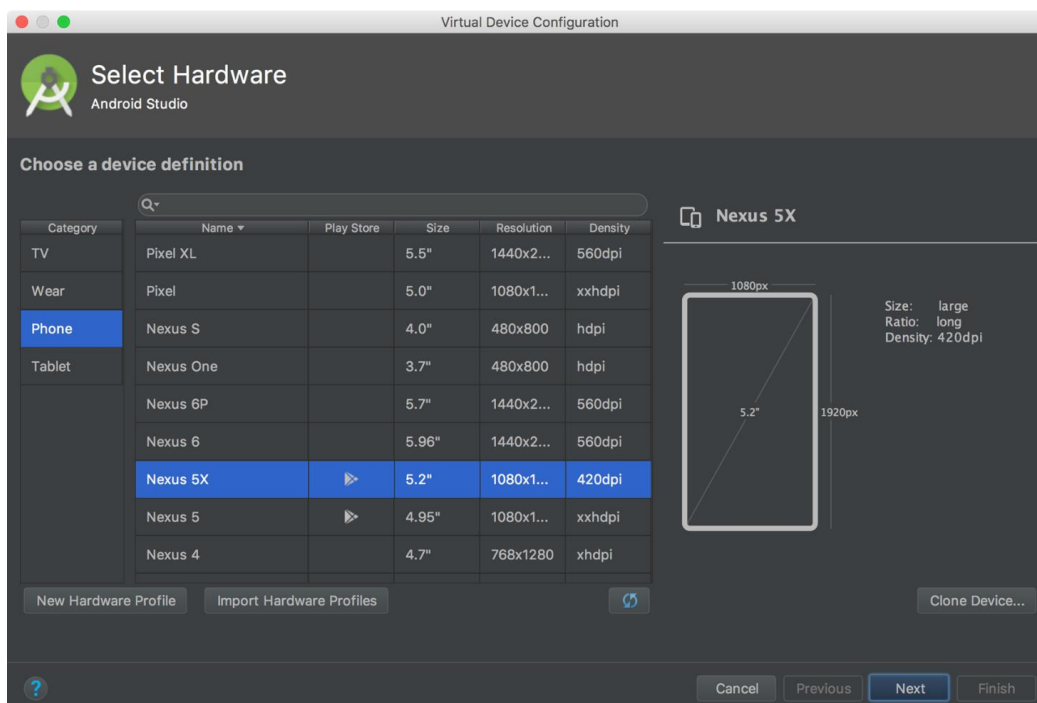


- Click AVD Manager in the toolbar.



Step 2:

- Click Create Virtual Device, at the bottom of the AVD Manager dialog.
- The Select Hardware page appears.



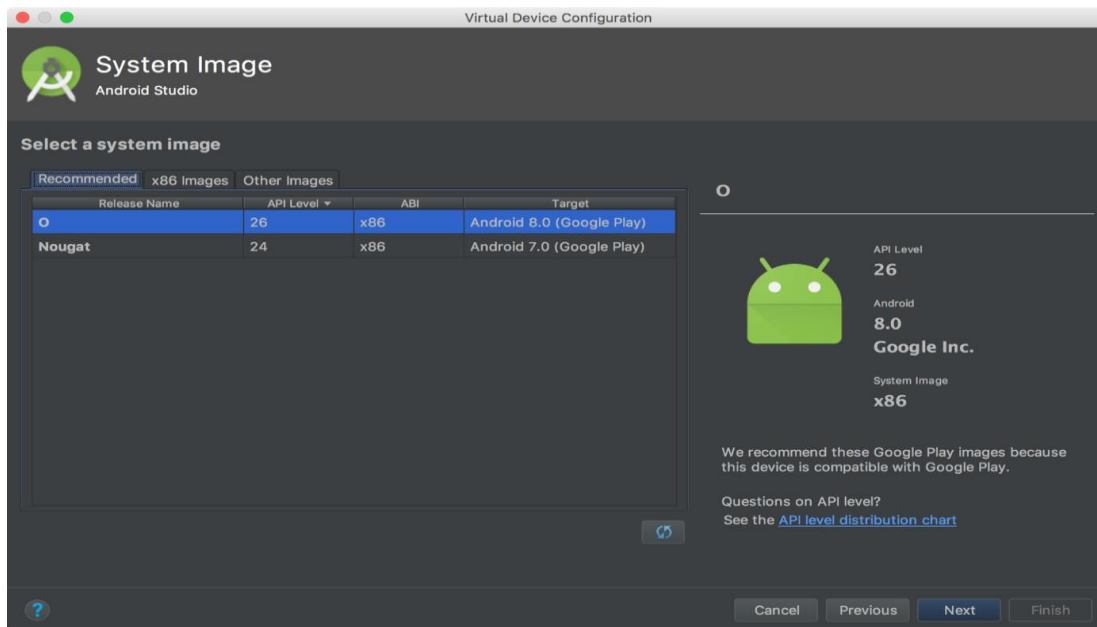
Step 3:

- Select a hardware profile, and then click Next.

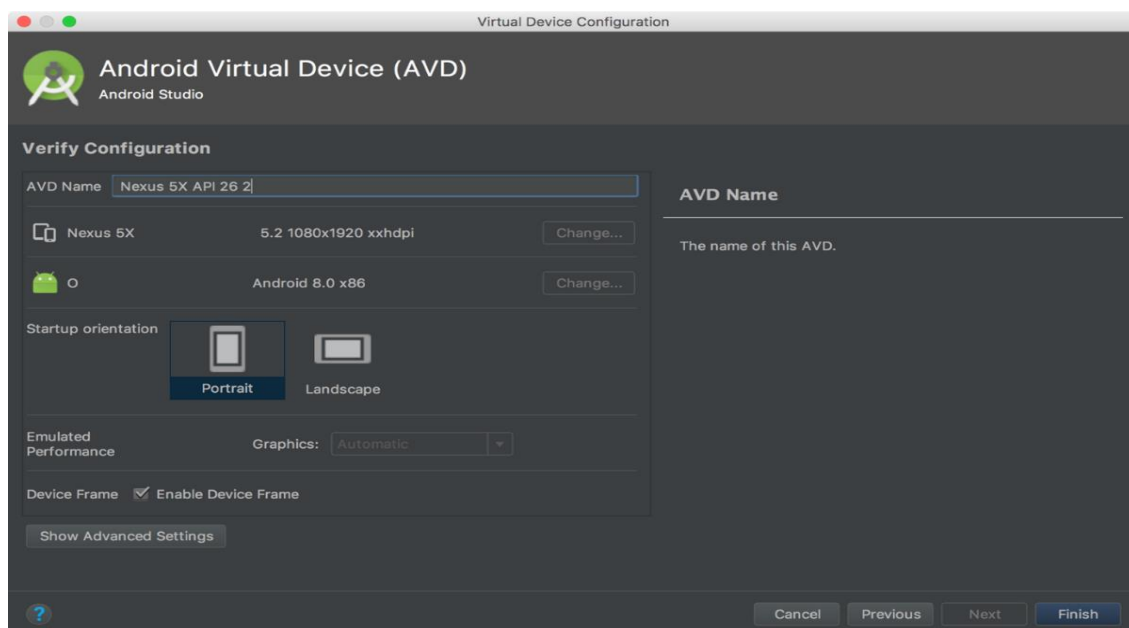
- If you don't see the hardware profile you want, you can create or import a hardware profile.

Step 4:

- The System Image page appears.



- Select the system image for a particular API level, and then click Next.

**Step 4:**

- Change AVD properties as needed, and then click **Finish**.

Step 5:

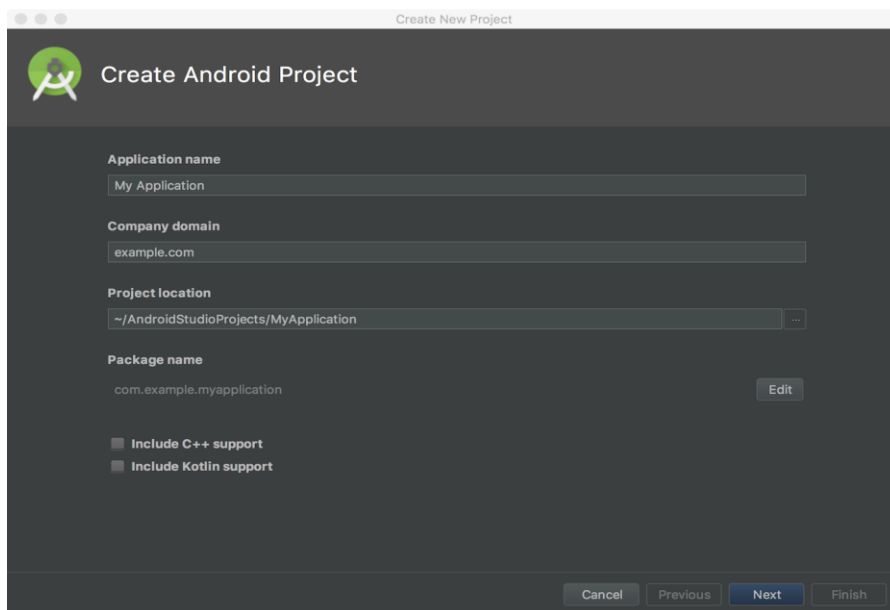
- The new AVD appears in the Your Virtual Devices page or the Select Deployment Target dialog.

14. Create an Android Studio Project

- Android Studio makes it easy to create Android apps for various form factors, such as handsets, tablets, TV, and Wear devices.

Step 1: Start and configure the project

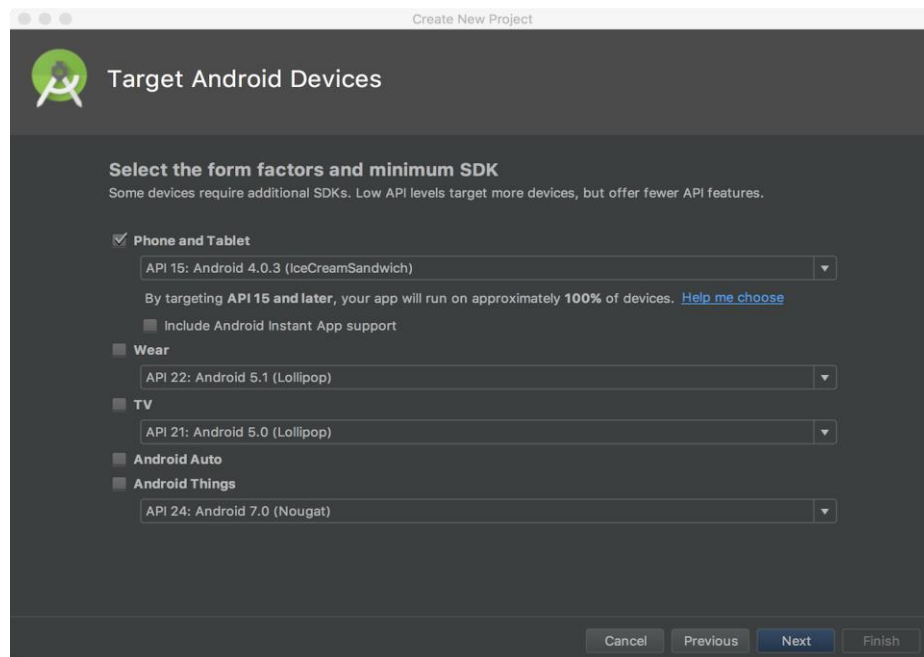
- Android Studio shows the Welcome screen, where you can click Start a new Android Studio project.
- If you do have a project opened, click File > New > New Project.
- In the Configure your new project screen that appears, you can set the name of your app, the package name, and the location of your project.



- Enter the values for your project then click Next

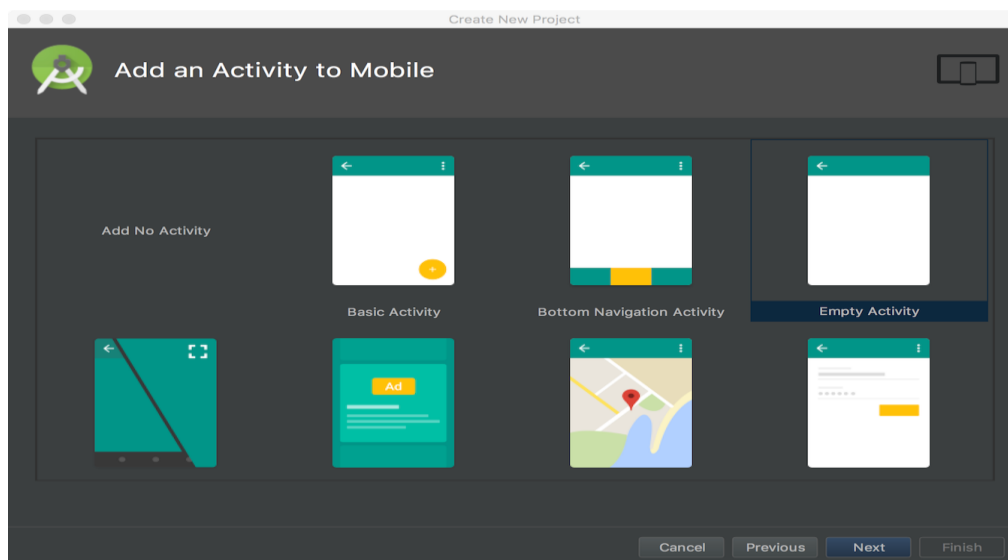
Step 2: Select form factors and API level

- The next window lets you select the device form factors you want to build for, and the minimum version



- Once you've selected your form factors and API versions, click Next

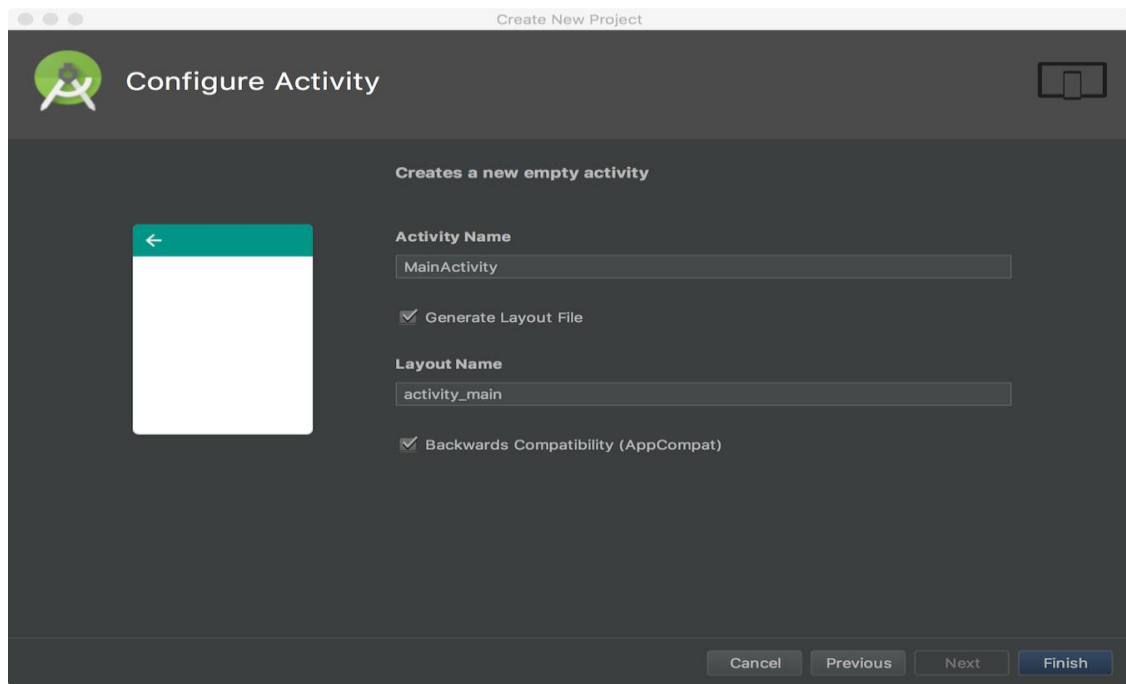
Step 3: Add an activity



- The next screen lets you select an activity type to add to your app
- Choose an activity type then click Next.

Step 4: Configure your activity

- The next screen lets you configure the activity to add to your app



- Enter the activity name, the layout name, and the activity title. Then click Finish. Android Studio now sets up your project and opens the IDE.
- Now you're ready to develop your app.

15. Hello world Program

MainActivity.java (program to create activity)

```
public class MainActivity extends Activity {  
  
    protected void onCreate(Bundle savedInstanceState) {  
  
        super.onCreate(savedInstanceState);  
  
        setContentView(R.layout.activity_main);  
  
    }  
  
}
```

Activity_main.xml

```
<android.support.constraint.ConstraintLayout>  
  
    <TextView  
  
        android:layout_width="wrap_content"  
  
        android:layout_height="wrap_content"
```

```
        android:text="Hello World!"

    />

</android.support.constraint.ConstraintLayout>
```

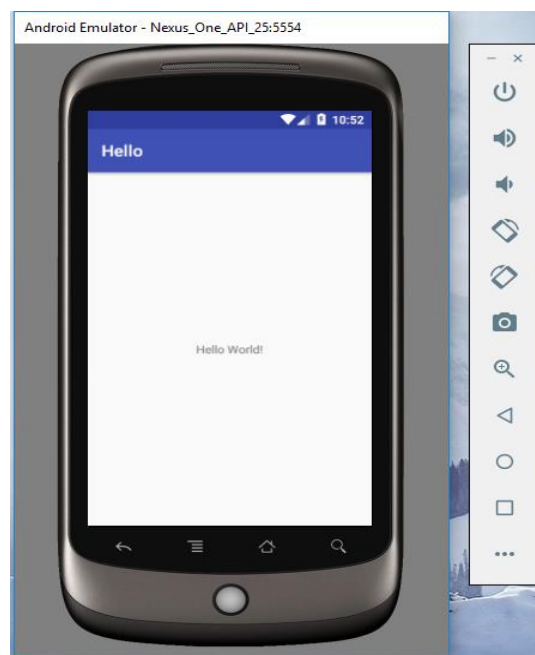
AndroidManifest.xml

```
<manifest>
    <application>

        <activity android:name=".MainActivity">
            <intent-filter>
                <action
android:name="android.intent.action.MAIN" />
                <category
android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>

        <activity android:name=".ActivityTwo">
        </activity>

    </application>
</manifest>
```



16. API (Application Programming Interface)

- API Level is an integer value that uniquely identifies the framework API revision

- Introduces a variety of new features and capabilities for users and developers.
- The Android platform provides a framework API that applications can use to interact with the underlying Android system.



17. List Various Mobile Communication Technologies

1G

- First Generation of wireless telephone technology
- 1G networks are analog telecommunication standard
- Information transfer rate 2.4 kbits/s

2G

- Difference between 1G and 2G, 2G networks are digital telecommunication standard
- Phone conversation are digitally encrypted
- 2G introduced mobile data, SMS and MMS Services
- 2G cellular telecom networks were commercially launched on the GSM standard
- Information transfer rate 50 kbits/s

3G

- Information transfer rate 3.1 Mbps
- Mobile internet, video calls and mobile TV
- New frequency band and high data rates

- 3.5G introduced LTE (Long Term Evolution)

4G

- Information transfer rate 100 Mbps
- Mobile broadband internet access
- HD mobile TV.
- Video conferencing
- Cloud computing

5G

- Availability of multiple data transfer rate
- 1Gbits/s data rate offered
- Coverage should be improved
- Signalling efficiency improved

Code Division Multiple Access (CDMA)

- Code Division Multiple Access (CDMA) is a sort of multiplexing that facilitates various signals to occupy a single transmission channel.
- It optimizes the use of available bandwidth.
- The technology is commonly used in ultra-high-frequency (UHF) cellular telephone systems, bands ranging between the 800-MHz and 1.9-GHz.
- Code Division Multiple Access system is very different from time and frequency multiplexing.
- In this system, a user has access to the whole bandwidth for the entire duration.
- The basic principle is that different CDMA codes are used to distinguish among the different users.
- Techniques generally used are direct sequence spread spectrum modulation (DS-CDMA), frequency hopping or mixed CDMA detection (JDCDMA).
- Here, a signal is generated which extends over a wide bandwidth. A code called **spreading code** is used to perform this action.
- Using a group of codes, which are orthogonal to each other, it is possible to select a signal with a given code in the presence of many other signals with different orthogonal codes.

EDGE (Enhanced Data for Global Evolution)

- EDGE (also known as Enhanced GPRS or EGPRS) is a data system used on top of GSM networks.
- It provides nearly three times faster speeds than the outdated GPRS system.
- The theoretical maximum speed is 473 kbps for 8 timeslots but it is typically limited to 135 kbps in order to conserve spectrum resources.
- Both phone and network must support EDGE, otherwise the phone will revert automatically to GPRS.
- EDGE meets the requirements for a 3G network but is usually classified as 2.75G.

Long Term Evolution (LTE)

- LTE is the next-step of the evolution of UMTS (3G) and HSDPA (3.5G).
- Some of the improvements LTE brings along over the currently used wireless mobile radio technologies are a better spectral efficiency, lower costs, higher transfer speeds, improved services, etc.
- Theoretically, LTE networks should provide wireless data *downlink* speeds of up to 300Mbps and *uplink* speeds of up to 75Mbps.
- Originally, LTE was conceived as an IP-based wireless system used purely for carrying data traffic.
- Network carriers were supposed to provide voice communication via their concurrent 2G/3G networks or by using VoIP.
- By popular request however, Voice over LTE (VoLTE) was a standardised system for transferring voice traffic over LTE.
- Currently, the availability of Voice over LTE (VoLTE) depends on the carrier implementation.
- Unfortunately, the current implementations of LTE use different frequency bands in different countries.

GSM (Global System for Mobile communication)

- GSM is a globally accepted standard for digital cellular communications.
- GSM uses narrowband Time Division Multiple Access (TDMA) for providing voice and text based services over mobile phone networks.
- GSM (Global System for Mobile communication) is a digital mobile network that is widely used by mobile phone users in Europe and other parts of the world.

- GSM uses a variation of time division multiple access (TDMA) and is the most widely used of the three digital wireless telephony technologies: TDMA, GSM and code-division multiple access (CDMA).
- GSM digitizes and compresses data, then sends it down a channel with two other streams of user data, each in its own time slot.
- It operates at either the 900 megahertz (MHz) or 1,800 MHz frequency band.

18. Files and Folders in Android Studio

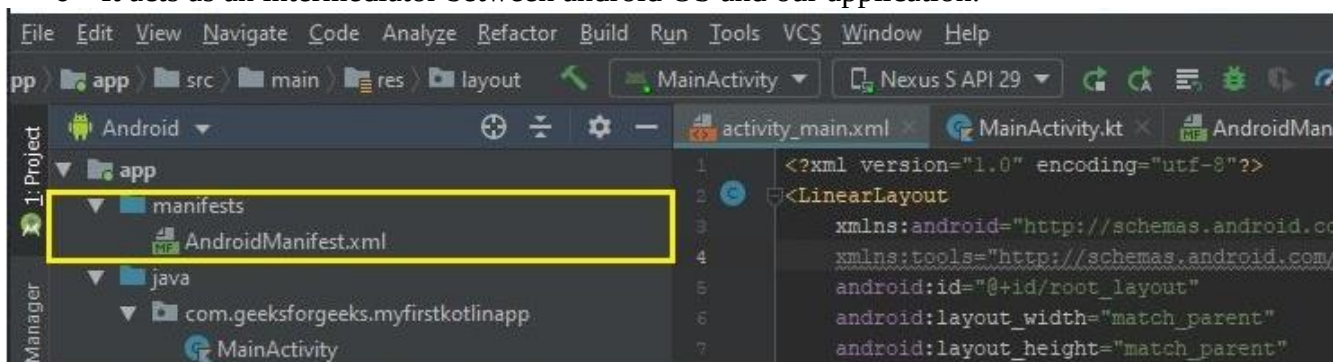
- The android project contains different type of app modules, source code files and resource files.
- We will explore all the folders and files in android app.

1. Manifests Folder
2. Java Folder
3. res (Resources) Folder
 - Drawable Folder
 - Layout Folder
 - Mipmap Folder
 - Values Folder

4. Gradle Scripts

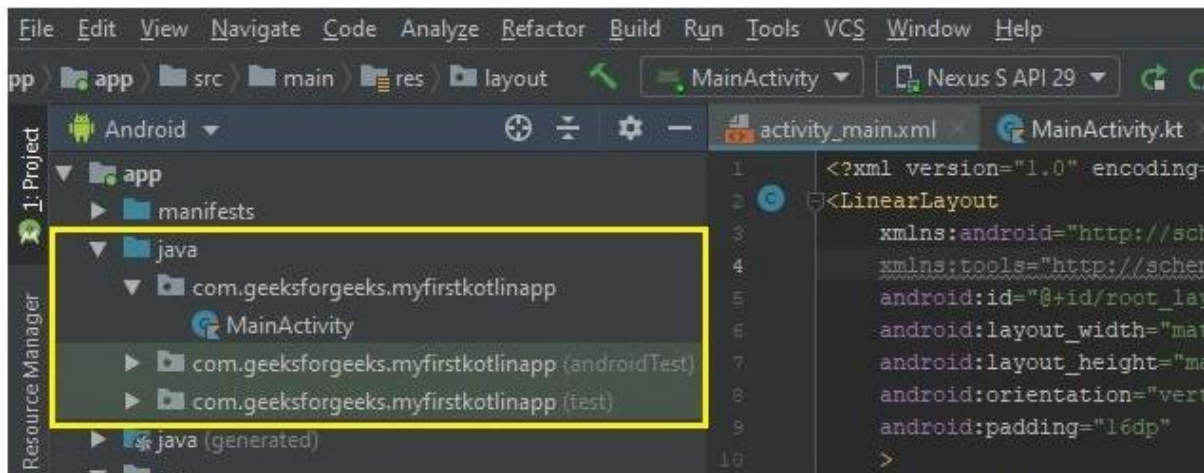
1. Manifests Folder

- Manifests folder contains **AndroidManifest.xml** for our creating the android application.
- This file contains information about our application such as android version, metadata, states package for Kotlin file and other application components.
- It acts as an intermediary between android OS and our application.



2. Java folder

- Java folder contains all the java and Kotlin source code (.java) files which we create during the app development, including other Test files.
- If we create any new project using Kotlin, by default the class file MainActivity.kt file will create automatically under the package name "com".



3.res/drawable folder

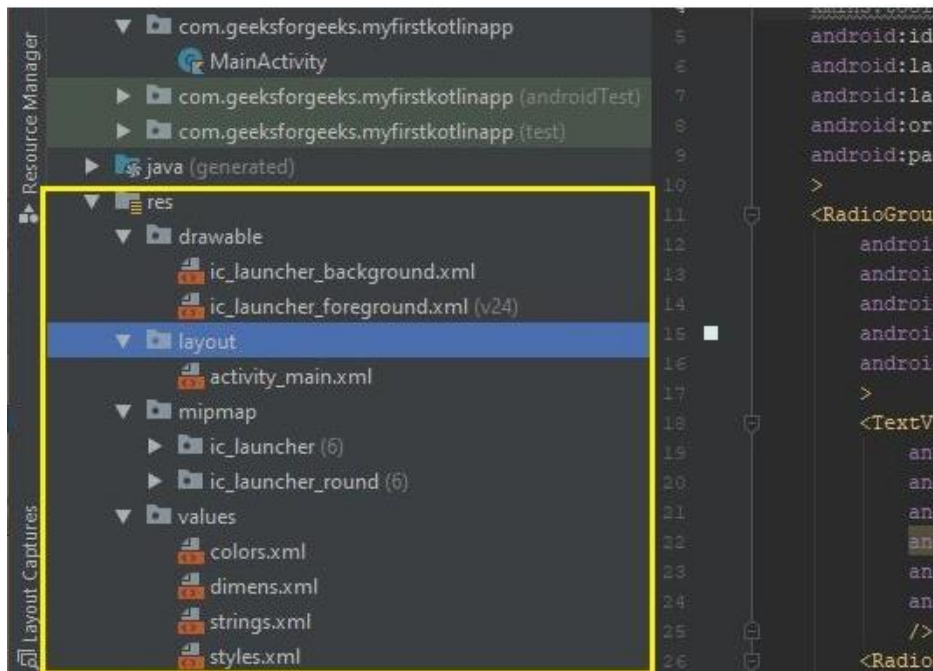
- It contains the different type of images used for the development of the application.
- We need to add all the images in drawable folder for the application development.

4.res/layout folder

- Layout folder contains all XML layout files which we used to define the user Interface of our application.
- It contains the **activity_main.xml** file.

5.Resource (res) folder

- Resource folder is the most important folder because it contains all the non-code sources like images, XML layouts, UI strings for our android application.



6.Gradle Scripts folder

- Gradle means automated build system and it contains number of files which are used to define a build configuration which can be apply to all modules in our application.

- In build.gradle (Project) there are buildscripts and in build.gradle (Module) plugins and implementations are used to build configurations that can be applied to all our application modules.

